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The Editor-in-Chief
Ocean Engineering
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Dear Editor,

Please find enclosed our manuscript titled **"WaveSplat: Grid-Free Spectral Gaussian Wave Field Estimation from Sparse Shipboard LiDAR"**, which we submit for consideration as a research article in *Ocean Engineering*.

The manuscript addresses a long-standing bottleneck in real-time shipboard wave sensing: grid-based ensemble Kalman filters (EnKF/ETKF), the de-facto standard for LiDAR and X-band wave field reconstruction, must interpolate sparse, irregularly distributed LiDAR returns onto a fixed spatial grid whose resolution is independent of the true scene complexity. We propose **WaveSplat**, a grid-free spectral representation in which the wave field is a weighted sum of N plane waves whose wavenumbers and amplitudes are directly optimised against sparse observations while the angular frequency is fixed by the linear dispersion relation as a hard constraint. An energy-based pruning and a residual-triggered growth mechanism make the number of components *adaptive*, matching scene complexity from one swell to multi-component seas.

Key contributions and findings:

- A grid-free spectral estimation framework with dispersion-constrained parameterisation and Fresnel-aware confidence weighting, runnable in real time on a single CPU thread.
- A pre-registered six-metric evaluation panel (nRMSE, variance ratio, Pearson ρ, Skill Score, directional R^2, H_s error) on 100 episodes spanning Beaufort 2–5 and 1–30 spectral components, compared against a hyperparameter-tuned ETKF-grid (24-config sweep) and a sliding-window GPR.
- WaveSplat attains **30.1% lower nRMSE** (0.223 vs. 0.319) and runs $\sim 10\times$ faster (13.4 vs. 132.6 ms/frame) than the best ETKF configuration, and is the *only* method with a positive Skill Score against a zero predictor.
- Improvements are highly significant over 100 paired runs (Wilcoxon $p{=}1.1 \times 10^{-15}$ vs. ETKF, paired Cohen's $d{=}1.16$; $p{=}4.7 \times 10^{-4}$ vs. sliding GPR, $d{=}0.42$), robust under two LiDAR raydrop models, and stable under 20–100% observation-density subsampling.

Relevance to Ocean Engineering. The paper sits squarely in the journal's scope of physical ocean measurement, data assimilation, and model-based sensing for marine operations. It directly enables downstream applications explicitly named in the journal's call, including wave-preview feedforward control for high-speed hydrofoil vessels (demonstrated in a companion paper), route optimisation, and structural-health monitoring.

Originality and prior communication. This work is original, has not been published previously, and is not under consideration elsewhere. A short early version of the grid-free idea was internally reviewed at Avikus but has not been submitted to any journal or conference. No

part of the manuscript has been previously published.

Data and reproducibility. The source code of WaveSplat, the physics-based LiDAR wave simulator (WaveSim), all experiment scripts, and five representative episodes are openly available at `https://github.com/SAEYONGPARK94/wavesplat`. The full 100-episode dataset can be regenerated from the provided configuration files and fixed random seeds, ensuring end-to-end reproducibility of every table and figure.

Suggested reviewers (no conflict of interest with the author):

- Prof. Nicolas Desmars — École Centrale Nantes — X-band/LiDAR ETKF reconstruction.
- Prof. Marc Buckley — Helmholtz-Zentrum Hereon — Wave field reconstruction from remote sensing.
- Dr. Jonas Ehlers — U-Net / FNO wave reconstruction.
- Prof. Ulrik Dam Nielsen — DTU — Ship-motion-based wave estimation.

The sole author declares no competing financial interests or personal relationships that could have influenced this work. The author has completed all listed CRediT roles (conceptualisation, methodology, software, validation, formal analysis, investigation, writing – original draft, writing – review & editing, visualisation).

Thank you very much for considering our manuscript. We look forward to the reviewers' comments.

Sincerely,

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